A Counterexample to the $o(s^{-2})$ Residual Rate for the Original Time-Scaled Cocoercive Newton Correction

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Abstract

Attouch, Boţ, and Nguyen prove that, for a cocoercive operator M with nonempty zero set, every solution of the Newton-corrected inertial dynamics

$$y''(s) + \frac{\alpha}{s}y'(s) + \frac{s}{\alpha+1}\frac{\mathrm{d}}{\mathrm{d}s}M(y(s)) + M(y(s)) = 0, \qquad \alpha > 1,$$

satisfies $\|M(y(s))\| = o(s^{-1})$ and $\int^{\infty} s\,\|M(y(s))\|^2\,\mathrm{d}s < \infty$. The same paper points to a closely related dynamics, with the coefficient $2s/(\alpha+1)$ in front of $\frac{\mathrm{d}}{\mathrm{d}s}M(y(s))$, for which the sharper estimates $\|M(y(s))\| = o(s^{-2})$ and $\int^{\infty} s^3\,\|M(y(s))\|^2\,\mathrm{d}s < \infty$ are known, and asks whether the same rate can be obtained by time-scaling and perturbation techniques.

This note gives a negative solution for the original coefficient $s/(\alpha+1)$, and therefore partial progress toward the broader modified-coefficient question. We construct a bounded linear 1/2-cocoercive operator M on a real Hilbert space, with $\operatorname{Zer} M = \{0\}$, and a classical solution of the original-coefficient equation such that

$$\|M(y(s))\|^2 \sim \frac{\sqrt{\pi}}{4}\big(2(\alpha+1)\big)^{3/2}s^{-3}.$$

Consequently $\|M(y(s))\| \neq o(s^{-2})$ and $\int^{\infty} s^3\,\|M(y(s))\|^2\,\mathrm{d}s = \infty$. The counterexample exploits an exact first-order subdynamics hidden in the coefficient $s/(\alpha+1)$; it does not rule out sharper estimates for genuinely modified equations such as the $2s/(\alpha+1)$ model.

1 Introduction and status of the result

Let H be a real Hilbert space. Recall that a single-valued operator $M : H \to H$ is ρ-cocoercive if, for some $\rho > 0$,

$$\langle Mx - My, x - y\rangle \geq \rho\,\|Mx - My\|^2 \qquad (x, y \in H). \tag{1}$$

Assume throughout that $\operatorname{Zer} M := \{x \in H : Mx = 0\}$ is nonempty. Attouch, Boţ, and Nguyen study, among other time-scaled dynamics, the operator equation

$$y''(s) + \frac{\alpha}{s}y'(s) + \frac{s}{\alpha+1}\frac{\mathrm{d}}{\mathrm{d}s}M(y(s)) + M(y(s)) = 0, \qquad \alpha > 1. \tag{2}$$

Their Theorem 13 proves the residual estimate

$$\|M(y(s))\| = o(s^{-1}), \qquad \int_{s_0}^{\infty} s\,\|M(y(s))\|^2\,\mathrm{d}s < \infty. \tag{3}$$

They also mention a closely related equation, in which the coefficient $s/(\alpha+1)$ in (2) is replaced by $2s/(\alpha+1)$, and for which the sharper estimates

$$\|M(y(s))\| = o(s^{-2}), \qquad \int_{s_0}^{\infty} s^3\,\|M(y(s))\|^2\,\mathrm{d}s < \infty \tag{4}$$

are available by other methods; see [1, Theorem 13 and Remark 10] and [2]. The question is whether the same convergence rate can be achieved using time scaling and perturbation techniques.

The result below answers one natural interpretation of that question negatively. Namely, for the exact equation (2), the estimates (4) are false under only the assumptions (1) and $\operatorname{Zer} M \neq \emptyset$. Thus the present note is a solution for the original-coefficient subproblem. It is not a solution of the broader modified-coefficient problem: it does not preclude a sharper rate for equations whose coefficients are genuinely changed, including the $2s/(\alpha+1)$ dynamics already treated in the literature.

Theorem 1 (Counterexample for the original coefficient)**.** *For every $\alpha > 1$ and every $s_0 > 0$, there exist*

(i) *a real Hilbert space H;*

(ii) *a bounded linear $1/2$-cocoercive operator $M : H \to H$ with $\operatorname{Zer} M = \{0\}$;*

(iii) *a classical solution $y : [s_0, \infty) \to H$ of (2);*

such that

$$\|M(y(s))\|^2 \sim \frac{\sqrt{\pi}}{4}\big(2(\alpha+1)\big)^{3/2} s^{-3} \qquad (s \to \infty). \tag{5}$$

Consequently

$$\|M(y(s))\| \neq o(s^{-2}) \quad \text{and} \quad \int_{s_0}^{\infty} s^3 \, \|M(y(s))\|^2 \, \mathrm{d}s = \infty.$$

The proof is constructive. The special coefficient $s/(\alpha+1)$ causes the second-order equation to contain an invariant copy of a first-order cocoercive flow. We choose a first-order flow whose residual decays like $t^{-3/4}$ in norm; after the time change $t = s^2/(2(\alpha+1))$, this becomes $s^{-3/2}$ in norm, which is too slow for (4).

2 The hidden first-order subdynamics

Set

$$\beta(s) := \frac{s}{\alpha+1}. \tag{6}$$

For bounded linear M, the left-hand side of (2) is

$$\begin{aligned} y'' + \frac{\alpha}{s} y' + \beta \frac{\mathrm{d}}{\mathrm{d}s} M(y) + M(y) \\ = \frac{\mathrm{d}}{\mathrm{d}s}\big(y' + \beta M(y)\big) + \frac{\alpha}{s}\big(y' + \beta M(y)\big), \end{aligned}$$

because

$$\beta'(s) + \frac{\alpha}{s}\beta(s) = \frac{1}{\alpha+1} + \frac{\alpha}{\alpha+1} = 1.$$

Thus every trajectory satisfying

$$y'(s) + \frac{s}{\alpha+1} M(y(s)) = 0 \tag{7}$$

is automatically a solution of the second-order equation (2).

Let

$$\tau(s) := \frac{s^2}{2(\alpha+1)}. \tag{8}$$

If z solves the autonomous first-order equation

$$\dot{z}(t) + M(z(t)) = 0, \tag{9}$$

and $y(s) := z(\tau(s))$, then

$$y'(s) = \tau'(s)\dot{z}(\tau(s)) = -\frac{s}{\alpha+1}M(y(s)),$$

so (7) holds. Therefore it remains to construct a cocoercive operator and a first-order solution whose residual has the desired slow asymptotic behavior.

3 The operator

Let

$$H := L^2\big((0,\pi);\mathbb{R}^2\big).$$

For $\theta \in (0,\pi)$, write

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

for the counterclockwise rotation by angle θ. Define the multiplication operator

$$(Mx)(\theta) := (I - R_\theta)x(\theta), \qquad x \in H. \tag{10}$$

Since $\|I - R_\theta\|_{\mathbb{R}^2\to\mathbb{R}^2} \le 2$, the operator M is bounded and linear on H.

Lemma 1. *The operator M in (10) is $1/2$-cocoercive and satisfies* $\operatorname{Zer} M = \{0\}$.

Proof. For every $v \in \mathbb{R}^2$ and every $\theta \in (0,\pi)$,

$$\begin{aligned} \langle (I-R_\theta)v, v\rangle_{\mathbb{R}^2} &= (1-\cos\theta)\,\|v\|_{\mathbb{R}^2}^2\,, \\ \|(I-R_\theta)v\|_{\mathbb{R}^2}^2 &= 2(1-\cos\theta)\,\|v\|_{\mathbb{R}^2}^2\,. \end{aligned}$$

Consequently, for $h \in H$,

$$\langle Mh, h\rangle_H = \frac{1}{2}\,\|Mh\|_H^2\,.$$

Since M is linear, applying this identity to $h = x - y$ yields

$$\langle Mx - My, x - y\rangle_H = \frac{1}{2}\,\|Mx - My\|_H^2\,,$$

which is exactly $1/2$-cocoercivity.

Finally, $I - R_\theta$ is invertible for every $\theta \in (0,\pi)$, because 1 is not an eigenvalue of a nontrivial planar rotation. Hence $Mx = 0$ implies $x(\theta) = 0$ for almost every θ, and therefore $\operatorname{Zer} M = \{0\}$. □

4 A first-order trajectory with slow residual decay

Let $u := (1,0) \in \mathbb{R}^2$ and set $z(0)(\theta) := u$. Because M is bounded linear, the first-order Cauchy problem (9) has the classical solution

$$z(t)(\theta) = \exp\big(-t(I - R_\theta)\big)u, \qquad t \ge 0. \tag{11}$$

Let

$$Q(t) := \|M(z(t))\|_H^2\,. \tag{12}$$

We compute $Q(t)$ explicitly. Put

$$J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$I - R_\theta = (1-\cos\theta)I - \sin\theta J.$$

The scalar part and the skew-symmetric part commute, and e^{aJ} is an isometry of $\mathbb{R}^2$. Therefore

$$\big\|\exp\big(-t(I-R_\theta)\big)u\big\|_{\mathbb{R}^2} = e^{-t(1-\cos\theta)}. \tag{13}$$

Moreover $I - R_\theta$ commutes with $\exp[-t(I-R_\theta)]$. Using the identities in the proof of Lemma 1, we obtain

$$Q(t) = \int_0^\pi 2(1-\cos\theta)e^{-2t(1-\cos\theta)}\,\mathrm{d}\theta. \tag{14}$$

Lemma 2. *As $t \to \infty$,*

$$Q(t) \sim \frac{\sqrt{\pi}}{4}t^{-3/2}. \tag{15}$$

Proof. Make the change of variables $x = \sqrt{t}\,\theta$ in (14). Extending the integrand by zero for $x > \pi\sqrt{t}$, we get

$$t^{3/2}Q(t) = \int_0^\infty F_t(x)\,\mathrm{d}x, \tag{16}$$

where

$$F_t(x) := \mathbf{1}_{[0,\pi\sqrt{t}]}(x)2t\left(1-\cos\frac{x}{\sqrt{t}}\right)\exp\left[-2t\left(1-\cos\frac{x}{\sqrt{t}}\right)\right].$$

For each fixed $x \geq 0$,

$$2t\left(1-\cos\frac{x}{\sqrt{t}}\right) \longrightarrow x^2,$$

and hence $F_t(x) \to x^2e^{-x^2}$.

For $0 \leq r \leq \pi$ we use the elementary estimates

$$1-\cos r \leq \frac{r^2}{2}, \qquad 1-\cos r \geq \frac{2r^2}{\pi^2}. \tag{17}$$

When $0 \leq x \leq \pi\sqrt{t}$, these give

$$0 \leq F_t(x) \leq x^2e^{-4x^2/\pi^2}.$$

The right-hand side is integrable on $[0,\infty)$. Dominated convergence in (16) yields

$$\lim_{t\to\infty} t^{3/2}Q(t) = \int_0^\infty x^2e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{4}.$$

This proves (15). $\square$

5 Proof of the main theorem

Fix $\alpha > 1$ and $s_0 > 0$. Let z be the first-order solution (11), and define

$$y(s) := z\left(\frac{s^2}{2(\alpha+1)}\right), \qquad s \geq s_0. \tag{18}$$

As explained in Section 2, y is a classical solution of (2). Indeed,

$$y'(s) = \frac{s}{\alpha+1}\dot{z}\left(\frac{s^2}{2(\alpha+1)}\right) = -\frac{s}{\alpha+1}M(y(s)),$$

so y satisfies (7), and hence (2).

By Lemma 2, with $\tau(s) = s^2/(2(\alpha+1))$,

$$\begin{aligned}
\|M(y(s))\|^2 &= Q(\tau(s)) \\
&\sim \frac{\sqrt{\pi}}{4}\left(\frac{s^2}{2(\alpha+1)}\right)^{-3/2} \\
&= \frac{\sqrt{\pi}}{4}\big(2(\alpha+1)\big)^{3/2} s^{-3}.
\end{aligned}$$

This is (5). Thus

$$s^2\,\|M(y(s))\| \sim \left[\frac{\sqrt{\pi}}{4}\big(2(\alpha+1)\big)^{3/2}\right]^{1/2} s^{1/2} \to \infty,$$

so $\|M(y(s))\|$ is not $o(s^{-2})$. Also

$$s^3\,\|M(y(s))\|^2 \longrightarrow \frac{\sqrt{\pi}}{4}\big(2(\alpha+1)\big)^{3/2} > 0,$$

and therefore

$$\int_{s_0}^{\infty} s^3\,\|M(y(s))\|^2\,\mathrm{d}s = \infty.$$

This proves Theorem 1. Notice that the example is still compatible with the known estimate (3), since $\|M(y(s))\| \asymp s^{-3/2} = o(s^{-1})$.

6 A strengthening: no uniform polynomial improvement over s^{-1}

The preceding example has residual norm of exact order $s^{-3/2}$. The same construction can be adjusted to show that, for the original coefficient, no estimate $o(s^{-1-\varepsilon})$ with a fixed $\varepsilon > 0$ follows from cocoercivity alone.

Proposition 1. *For every $\alpha > 1$, every $s_0 > 0$, and every $\varepsilon > 0$, there exist H, M, and a classical solution y of (2), with M bounded linear and $1/2$-cocoercive and* $\operatorname{Zer} M = \{0\}$*, such that*

$$\|M(y(s))\| \neq o(s^{-1-\varepsilon}).$$

Proof. Use the same H and M, but choose a singular initial datum

$$z_b(0)(\theta) := \theta^{-b/2}u,$$

where $u = (1,0)$ and $b \in (-1,1)$. This datum belongs to H exactly because $b < 1$. The first-order solution is again

$$z_b(t)(\theta) = \exp[-t(I - R_\theta)]\theta^{-b/2}u,$$

and its residual satisfies

$$Q_b(t) := \|M(z_b(t))\|^2 = \int_0^{\pi} 2(1-\cos\theta)\theta^{-b}e^{-2t(1-\cos\theta)}\,\mathrm{d}\theta. \tag{19}$$

The same scaling $x = \sqrt{t}\,\theta$, now with the integrable majorant

$$C_b\,x^{2-b}e^{-4x^2/\pi^2}$$

near infinity and integrability near zero ensured by $b < 1$, gives

$$Q_b(t) \sim \frac{1}{2}\Gamma\left(\frac{3-b}{2}\right)t^{-(3-b)/2}. \tag{20}$$

After the time change $t = s^2/(2(\alpha+1))$, the resulting solution of (2) satisfies

$$\|M(y_b(s))\| \asymp s^{-(3-b)/2}.$$

Choose $b < 1$ sufficiently close to 1 that $(3-b)/2 < 1+\varepsilon$. Then

$$s^{1+\varepsilon}\,\|M(y_b(s))\| \to \infty,$$

so $\|M(y_b(s))\|$ is not $o(s^{-1-\varepsilon})$. $\square$

7 Conclusion

The estimates

$$\|M(y(s))\| = o(s^{-2}), \qquad \int_{s_0}^{\infty} s^3\,\|M(y(s))\|^2\,\mathrm{d}s < \infty$$

are false for the original Newton-corrected cocoercive-operator dynamics (2). Hence the original-coefficient version of the question has a negative answer under the stated assumptions. The obstruction is structural: the coefficient $s/(\alpha+1)$ exactly embeds a time-rescaled first-order cocoercive flow, and first-order cocoercive flows can have arbitrarily slow polynomial improvement beyond the $t^{-1/2}$ residual scale in norm.

This is partial progress, not a full settlement of the broader question about modified coefficients or modified time-scaling/perturbation frameworks. In particular, the argument does not challenge the known positive results for the coefficient $2s/(\alpha+1)$ or for Fast-OGDA-type Newton corrections; it only shows that those stronger estimates cannot be true for the exact equation (2) under cocoercivity alone.

References

[1] H. Attouch, R. I. Boţ, and D.-K. Nguyen, *Fast Convex Optimization via Time Scale and Averaging of the Steepest Descent*, Mathematics of Operations Research 50(4):2633–2665, 2025. https://doi.org/10.1287/moor.2023.0186

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[3] Open Problems in Operations Research, *Achieve faster residual decay for cocoercive-operator Newton-corrected inertial dynamics*, problem ID W4403453300_p0. https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4403453300_p0&page=12&n=135